Warren Hsu

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Education

University of Illinois at Urbana-Champaign
BS in Mathematics & Computer Science, Minor in Philosophy — Expected: May 2029

Awards: Campus Honors Program, James Scholar Honors Program, 1600 SAT, Seal of Biliteracy in Chinese

Languages: Python, SQL, R, C, C++, JavaScript, Kotlin

Skills: Git/Github, scikit-learn, XGBoost, Pytorch, TensorFlow, pandas, NumPy, Tableau, Google Earth Engine, GeoPandas, OpenCV, matplotlib, Jupyter Notebook, Google Cloud Platform, API Integration

Experience

Climate Modeling, Predictions, and Applications Group - Research Assistant — Oct. 2025—Current

- Developed neural operator models for atmospheric prediction using idealized simulations, optimizing training pipeline with **50%** memory reduction and deploying on university supercomputer via batch job scheduling
- Built diagnostic toolkit to validate simulation accuracy through physics-informed error metrics, testing model performance across **50** initial conditions with multi-step rollout validation
- Generated atmospheric dataset using spectral methods with comprehensive visualization framework (sphere projections, spectral analysis), enabling machine learning research on atmospheric modeling with physical consistency

Statistics in the Community - Data Analyst — Sep. 2025—Current

- Analyzed 25,000+ food assistance program records to identify unique participants by removing duplicate entries, enabling accurate demographic reporting for Eastern Illinois Foodbank
- Built analytics pipeline in Python and R to generate demographic visualizations for client presentations, showing gender and race/ethnicity distributions across program participants
- Automated demographic reporting to track program equity and support resource allocation decisions for community food assistance initiatives

Global Trade Capital Foreign Exchange Ltd. - Software Development Intern — Jun. 2024—Aug. 2025

- Engineered Python and SQL pipelines to analyze broker login data for over **100,000** international clients
- Designed automated notification system identifying over **5000** security threats through data analysis
- Utilized cloud-based data tools and Github version control to enhance reliability and collaboration

Florida State University - Research Assistant — Jun. 2024—Aug. 2024

- Researched possibility theory under Professor Daniel G. Schwartz from Florida State University
- Implemented Dijkstra's algorithm in Python for pathfinding to compare possibility and probability theory
- Contributed to a paper currently under review titled *A Concept of Possibility for Real-World Events* Paper

Projects

Image-based ML Covid Predictor - Machine Learning Engineer — Oct. 2025—Dec. 2025

- Built ML models improving COVID-19 prediction accuracy by **1,300%** across **1,278** US counties, processing satellite imagery from **1,479** counties with **89.5%** matching accuracy
- Demonstrated satellite imagery can replace traditional housing data in pandemic prediction models while maintaining **95%** accuracy, enabling analysis in areas lacking census data Github

Independent Applied Mathematics Paper - Researcher — Jun. 2024—Oct. 2024

- Authored research paper on possibility theory for stock market indicators, developing 11 simulations using Python and SQLite with 24 years of data; viewed 156 times on TechRxiv Paper Github
- Advised by Professor Daniel G. Schwartz from Florida State University